Syed Emad Uddin Shubha

syed.shubha@ieee.org | in/syedshubha | syedshubha.github.io

Citizenship: Bangladesh , US Visa Status: F-1 Student

RESEARCH INTEREST

- Quantum Communication
- Quantum Hardware Security
- Quantum Cryptography
- Quantum Algorithm
- Quantum Error Correction
- Deep Learning

EDUCATION

Louisiana State University **January 2025 – Present**
Computer Science Graduate Student **CGPA: 4.00/4.00**
Relevant Courses: Quantum Technology, Algorithm Design and Analysis, Machine Learning.

Bangladesh University of Engineering and Technology **February 2017 – May 2022**
B.Sc. in Electrical and Electronic Engineering **CGPA: 3.24/4.00**
Relevant Courses: Control Systems, Solid State Devices, Random Signals and Processes, Wireless Communication.

RESEARCH EXPERIENCE

Graduate Research Assistant — Louisiana State University, USA January 2025 - Present

- Characterized crosstalk as an active attack vector through initial pulse-level simulations (QCE 2025); developed this work into a full pulse-to-circuit framework that identified a novel asymmetric attack mechanism (ACM QSec 2025).
- Investigated the role of quantum correlations in dense coding and the design of an adaptive purification protocol. Drafted research aims, questions, and methodology for an NSF EAGER proposal (prior approval secured based on preliminary results presented at QCE 2025).

Research Assistant — North South University, Dhaka June 2022 - May 2024

- Pioneered Bangladesh's first research initiatives in quantum computing and communication. Extended my undergraduate thesis (where I developed a simulation framework to analyze the fidelity of encoded Bell states under noise) into a first-authored high-impact journal article, proposing a novel error correction protocol that uses stabilizer formalism to improve fidelity under Pauli noise channels for both local and long-distance communication.
- Mentored junior researchers; formulated and directed research projects. Research on surface code decoder design was presented at an IEEE conference; integrated QKD with classical security, resulting in two additional journal articles.

JOURNAL PUBLICATIONS

1. "Secure and Efficient n-Qubit Entangled StateTeleportation Using Partially Entangled GHZ Channels and Optimal POVM." (**Corresponding author**; accepted in AVS Quantum Science, Q1-indexed, IF: 3.0), **doi: 10.1116/5.0284072**.
2. "Significant improvement of fidelity for encoded quantum bell pairs at long and short-distance communication along with a generalized circuit." (**1st author**; accepted in Heliyon, Q1-indexed, IF: 3.6), **doi: 10.1016/j.heliyon.2023.e19700**.
3. "Multi-Layered Security System: Integrating Quantum Key Distribution with Classical Cryptography to Enhance Steganographic Security". (Co-supervised, accepted in Alexandria Engineering Journal, Q1-indexed, IF=6.8), **doi: 10.1016/j.aej.2025.02.056**.
4. "Enhancing the security of image transmission in Quantum era: a chaos-assisted QKD approach using entanglement." (Co-supervised, accepted in IET Quantum Communication, Q2-indexed, IF=2.8), **doi: 10.1049/qtc2.70016**.

SELECTED PEER-REVIEWED CONFERENCE PROCEEDINGS

5. "Pulse-to-Circuit Characterization of Stealthy Crosstalk Attack on Multi-Tenant Superconducting Quantum Hardware" (**1st author**). Accepted for publication in ACM Quantum Security Workshop 2025, **doi: 10.1145/3733825.3765282**.
6. "Testbed Protocol for Adaptive Superdense Coding in Distributed Quantum Networks" (**1st author**). Presented at IEEE Quantum Week, 2025. Forthcoming in IEEE Xplore. **arXiv: 2504.12565**.
7. "Pulse-Level Simulation of Crosstalk Attacks on Superconducting Quantum Hardware" (**1st author**). Presented at IEEE Quantum Week, 2025. Forthcoming in IEEE Xplore. **arXiv: 2507.16181**.

TEACHING EXPERIENCE

Graduate Teaching Assistant — Louisiana State University, USA — August 2025 - Present

- Supporting the instructional activities for CSC 4501 (Computer Networks)
- Grading assignments and providing feedback to students.

Teaching Assistant — BRAC University, Dhaka — June 2023 - December 2024

- Conducted CSE 482 (Quantum Computing II) tutorials and problem-solving classes.
- Conducted lab sessions of MAT120 (Integral Calculus and Ordinary Differential Equations), teaching relevant numerical methods and Python libraries.

NOTABLE PROJECTS

- **Preparing the W State on Superconducting Qubits Using QISKIT PULSE** — ***GitHub Link***.
  - Used real calibration data of ibm_sherbrooke to design necessary gate and measurement pulses to prepare W states.
  - Performed Rabi Experiment and Eco-CR calibration to adjust pulse amplitudes.
- **Quantum Image Encoding and Decoding using QISKIT** — ***GitHub Link***.
  - Implemented quantum image encoding algorithm using FRQI and NEQR method.
  - Generalized both encoding circuit, developed the decoding method and reconstructed the output image.
- **Load Flow Analysis for n Bus using Gauss-Siedel Algorithm** — ***GitHub Link***.
  - Developed a load flow analysis program for n-bus power systems during the junior year of undergraduate studies.
  - Implemented generalized Gauss-Siedel algorithm and generated output for user defined tolerance.

SKILLS SUMMARY

Programming Language	: Python, MATLAB, C, C++, Assembly language.
Simulation Software	: PSpice, Proteus, Quartus, Simulink.
Office Application	: Microsoft Office Suite, LaTeX.
Soft Skills	: Organizing, Collaborating, Teaching, Writing.

HONORS & AWARDS

- Recipient of the **NSF Student Travel Grant** ($1250), to present two first-author papers at QCE 2025
- Presented at Q-Net Symposium by LSU APS Chapter (2025).
- Invited Speaker at SQuIC, BRACU on Quantum Communication (2024).
- Instructor at the 3^{rd} PSI-Tensor Winter School for theoretical physics (2024).
- Keynote Speaker for an IEEE NSU Student Branch event on Quantum Computing (2022).
- First Place, University-Level Math Olympiad, organized by Bangladesh Mathematical Society (2021).
- Technical Scholarship (up to all semesters) at BUET (2017 to 2022).
- Ranked 47th among 10,000 candidates in the BUET Admission Test, earning University Admission Test Scholarship (2017).

RECOGNITION & MEDIA COVERAGE

- Research published in ACM QSec 2025 was featured in the industry publication Quantum Zeitgeist — **Link**.
- Recognized by the LSU Division of CSE for research contributions at IEEE Quantum Week 2025 — **Link**.

SELECTED CERTIFICATIONS

- Completed "Tensor PSI Winter School for Theoretical Physics". (2023)
- Completed "Qiskit Global Quantum Summer School" organized by The Coding School. (2021)

LEADERSHIP & OUTREACH

- Assisted in the distribution of PPE during the COVID-19 emergency.
- Volunteered in the distribution of winter clothes in remote villages.
- Organized a nationwide online course on quantum mechanics and quantum computing.
- Engaged in numerous science outreach initiatives through writing blogs and articles.